AGRICULTURAL DRAINAGE · NOTE DR04

Subsurface Drainage Principles — Tile Systems, Darcy Flow, Drain Inflow and the Auger-Hole Test

Course topics: IDE-502 T5, IDE-502 T7, IDE-506 T2

Purpose of subsurface drainage

A subsurface drainage system controls the water table and keeps a favourable salt balance in the crop root zone. It may use buried horizontal pipelines, deep open ditches or drainage wells, alone or together. Deep open drains mainly remove excess surface water and act as outlets for underground pipe (tile) drains.

Type	Details
Horizontal — tile drainage	A network of buried pipes: laterals → collectors → open-ditch outlet. Burnt-clay or cement-concrete tiles 10–15 cm in diameter and 30, 60 or 65 cm long; PVC pipe comes in continuous lengths of 200–250 m. Laterals are laid 1–1.5 m deep and 30–60 m apart. Collectors are deeper, of the same material but larger and non-perforated
Horizontal — mole drains	Cylindrical or oval underground earth channels formed in cohesive or fibrous organic soils; limited application
Vertical	A system of drainage wells

How a tile system works

- Tiles are laid in a continuous line at a set depth and grade so free water enters and flows out by gravity. Components: drainage outlet, collector drains and laterals — laterals take free water from the soil; collectors carry it to the outlet.
- Purpose: lower the groundwater below the root zone, removing only excess gravitational water — not water plants can use. Path: soil → laterals → collectors → outlet.
- Compared with surface drains, subsurface drains take no land, harbour no weeds and do not hinder farming; most fields need a combination of surface and subsurface drainage.

Advantages

- Better aeration and soil temperature; a larger effective root zone for nutrient uptake; more bacterial activity, better structure and nutrient availability; more available moisture in the root zone.
- Less erosion — the soil holds more rain and sheds less runoff; better moisture conditions for tillage, planting and harvest; toxic substances (e.g. alkali) removed from the root zone.
- **Roots:** on undrained land the high free-water level restricts roots; on tile-drained land the lower water level gives a deeper, healthier root system.
- **History:** early drains were narrow trenches filled with poles, brush, stones or wooden troughs; they silted up or collapsed as the wood rotted and worked only a few years. Modern practice uses tiles.

Flow in the saturated zone

Flow in saturated soil involves mechanical, chemical and thermal energy and molecular attraction; only the mechanical forces are considered here. Total energy E = kinetic + pressure + elevation energy, but groundwater moves so slowly that kinetic energy is negligible. The flow potential is the **hydraulic head**:

$$H = p/w + Z$$

H = hydraulic head (m); p = pressure relative to atmosphere (kg/m^2); w = specific weight of water (kg/m^3); Z = elevation above datum (m)

- At the water table p/w = 0, so H = Z. Below the ground, measure the head with piezometers.
- **Permeability** is the capacity of a porous medium to transmit fluids — in drainage, the property that lets water move through saturated soil under hydrostatic pressure. **Hydraulic conductivity** is its numerical value: the coefficient k in Darcy's law.
- **Effective velocity** is the flow per unit time divided by the total area of the porous medium (not just the voids).

$$\text{Darcy's law: } v = k \cdot i$$

$$\text{Rate of flow: } Q = a \cdot v = a \cdot k \cdot i$$

$$\text{Hydraulic gradient: } i = (H_1 - H_2)/L = \{[(p_1/w) + z_1] - [(p_2/w) + z_2]\}/L$$

- v = effective velocity; k = coefficient of permeability; i = hydraulic gradient; a = flow area; L = length of flow path; subscripts 1 and 2 mark the higher and lower head. Units of k are usually cm/hr or m/day. The law holds for almost all natural drainage situations.
- If i = 1 then v = k, so k has the dimensions of velocity (L/T). Head difference drives groundwater as temperature difference drives heat and voltage drives current.
- **What raises k:** high porosity, coarse open texture and well-developed structure; a few large pores carry more than many small ones, and fine soils depend on structural pores. Water quality (salinity, alkalinity) also affects k. Soils may be non-homogeneous because of stratification, barriers and direction-dependent conductivity. **Horizontal hydraulic conductivity** measures horizontal movement.
- Q = a·k·i serves simple estimates; where gradient and conductivity vary, indirect methods are needed.

Term	Meaning
Porosity	Volume of voids ÷ total soil volume. (Percolation is the downward movement of soil water.)
Drainable porosity (specific yield)	Volume of water drained by gravity and soil-water tension ÷ total saturated soil volume × 100
Drainage coefficient (drainage modulus)	Design rate of water removal — depth removed per unit time, usually 24 hours
Drainage efficiency	Water discharged by drains ÷ precipitation in the same period
Steady flow	Flow whose discharge per unit time stays constant

Movement of water into drains

- **Factors:** hydraulic conductivity of the soil horizons; the position of the free water surface, artesian pressure and back pressure in the drain; drain depth and position relative to the horizons; drain

spacing and diameter; tile-joint or PVC hole spacing; depth to the impervious layer.

- **Entry:** water enters through the cracks between pipes or through holes — little or none through the walls. A mole drain takes water over its whole permeable perimeter and through the open wedge above it.
- The soil right around the drain controls the flow, and most head loss happens there. Flow rate relates to the rate of fall of the water table and the drain spacing.
- Between tile lines the water table is **semi-elliptical**; its movement depends on the size and number of pores.
- **Fine soils** percolate slowly. Heavy clays have small pores and colloidal clogging and little removal area, so economic spacing limits tile use. **Sandy soils** move water fast and often drain naturally; they need artificial drainage mainly for a high water table or salt build-up, and sand may enter the tiles and reduce their effectiveness.

Hydraulic conductivity and its measurement

- Drain and well design depends on how well soils and aquifers transmit water. "Hydraulic conductivity", "permeability" and "coefficient of permeability" are often used as synonyms, though they differ: hydraulic conductivity is k in v = k·i; permeability is the soil's ability to transmit water; the coefficient of permeability is the flow through unit area under unit gradient. k depends on both fluid and medium.
- **Laboratory:** samples (undisturbed cores from a core sampler, or disturbed soil dried, granulated and packed in a tube) are put in a permeameter under a known head for a known time; the water passed gives k. Transport disturbs samples and changes k, so laboratory values are **not suitable for tile drainage design** — measure in the field. Field methods differ with the presence or absence of a water table.

Auger-hole method (Hooghoudt single auger hole)

The commonest field method where a water table is present. It mainly measures the **horizontal** component of k — the one that matters for drainage. The formulas come from exact or approximate solutions of the Laplace equation.

1. Bore a hole below the water table and let the water level reach equilibrium with the soil water.
2. Pump the hole out to a new level.
3. Measure the rate of rise and compute k from it.

- There are two cases: the hole reaches the impervious layer, or the impervious layer is far below the hole bottom.
- **Assumptions:** the water table around the hole is not lowered by pumping (roughly true for a short time after pumping); water flows horizontally into the sides and vertically up through the bottom.

$$\text{Rise from side flow: } dy/dt = -k\cdot(2\pi ad/\pi a^2)\cdot y/S, \text{ with } S = ad/0.19 \text{ (empirical)}$$
$$\text{Rise from bottom flow: } dy/dt = -k\cdot y/S$$
$$\text{Combined: } dy/dt = -k\cdot(2d + a)\cdot y/(aS)$$
$$\text{Integrated: } \ln(y_0/y_1) = k(2d + a)\Delta t/(aS) \rightarrow k = [2.3\cdot aS/((2d + a)\Delta t)]\cdot\log_{10}(y_0/y_1)$$
$$\text{Hole ending on the impervious layer: } k = [2.3\cdot aS/(2d\cdot\Delta t)]\cdot\log_{10}(y_0/y_1)$$
$$\text{With } S = ad/0.19\text{: } k = 5{,}23{,}000 \cdot a^2\cdot\log_{10}(y_0/y_1)/\Delta t \text{ (a in m, } \Delta t \text{ in s, k in m/day)}$$

a = hole radius; d = depth of the hole below the water table; y = depth of the water level in the hole below the water table; S is a shape factor (not drain spacing) that depends on a, d, s and the water height in the hole. The coefficient 0.19 is accurate within about 27% in the field. In-situ k ranges from 0.001 to more than 10 m/day. Check of the constant: 2.3/(2 × 0.19) = 6.05 per second; × 86,400 s/day ≈ 5,23,000.

> **Example (as given in the notes)**
> y_0 = 90 cm, y_1 = 87 cm, a = 3 cm = 0.03 m, Δt = 19 s
> $k = 5{,}23{,}000 \times (0.03)^2 \times \log_{10}(90/87) / 19 = 5{,}23{,}000 \times 0.0009 \times 0.01472 / 19 \approx 0.36$ m/day
> The notes print the answer as k = 35.8 m/day.

Correction: with the notes' own formula and data, k ≈ 0.36 m/day (about 36 cm/day). 35.8 m/day is about 100 times too large and above the stated field range (> 10 m/day is already very high), so the printed value is most likely 35.8 ***cm****/day.*

Field procedure

- **Hole:** an unlined hole 7.5–10 cm in diameter, extending below the water table, if the walls are stable. In unstable soil or quicksand, line it with perforated pipe — drill, insert the pipe, drive it deeper and remove the soil. Do not compact the walls.
- Leave the hole for several hours to reach the static water table; measure the static depth with an electrical depth gauge or chalked tape.
- Bail the water out quickly to the required depth. At once lower the gauge to touch the water surface and note the reading and time; raise the probe a fixed distance (~2 cm) and note the time when the rising water touches it and completes the circuit. Repeat for equal successive rises until the water has risen about one-third of the drawdown.
- Repeat the whole test 2–3 times for accuracy.
- **Mapping:** measure k at several grid points and draw iso-conductivity lines like contours — the map of k variation guides tile drainage design.

Sources: ICAR eCourse 'Drainage Engineering' — Dr M. K. Jha (IIT Kharagpur), reviewed by Prof. K. Yellareddy (Director A&R, WALAMTARI, Hyderabad); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Darcy's law states that

(a) v = k/i (b) v = k·i
(c) v = k·i² (d) Q = k/a

Q2. At the water table the hydraulic head equals

(a) zero (b) the pressure head
(c) the elevation head Z (d) p/w + 2Z

Q3. The auger-hole method mainly measures

(a) vertical k (b) horizontal k
(c) unsaturated k (d) intrinsic permeability

Q4. Laboratory permeameter values of k are unsuitable for tile design mainly because

(a) they are too slow (b) samples are disturbed
(c) water is too pure (d) they ignore temperature

Q5. Between parallel tile lines the water table is roughly

(a) flat (b) triangular

(c) semi-elliptical (d) parabolic upward

Q6. In the auger-hole formula $k = 5{,}23{,}000 \cdot a^2 \cdot \log(y_0/y_1)/\Delta t$, a and Δt are in

(a) cm and min (b) m and s

(c) mm and h (d) m and h

Q7. Drainable porosity is also called

(a) effective porosity of clay (b) specific yield

(c) void ratio (d) field capacity

Q8. Lateral tile drains are usually laid at depths of

(a) 0.3–0.5 m (b) 1–1.5 m

(c) 2–3 m (d) 4–5 m

Answers

Q1 — (b) v = k·i

Q2 — (c) the elevation head Z — *p/w = 0 at the water table.*

Q3 — (b) horizontal k

Q4 — (b) samples are disturbed

Q5 — (c) semi-elliptical

Q6 — (b) m and s — *k then comes out in m/day.*

Q7 — (b) specific yield

Q8 — (b) 1–1.5 m